CP255 - W12_S1 Debugging Answer Key

Your turn: Exercise 1: Read the error

Run each cell below. For each one, write a comment explaining:

- What type of error it is
- What caused it
- How to fix it

Then fix the code.

Python

```
# Bug A: run this, read the error, then fix it
file_counts = {"csv": 12, "pdf": 8, "docx": 5}
print("Number of xlsx files:", file_counts["xlsx"])
```

Python

```
# Error Type: KeyError
# Cause: "xlsx" is not a key in the dictionary
# Fix: use an existing key or use .get()

file_counts = {"csv": 12, "pdf": 8, "docx": 5}

print("Number of xlsx files:", file_counts.get("xlsx", 0))
```

Python

```
# Bug B: run this, read the error, then fix it
report_numbers = ["01", "02", "03"]
last_report = report_numbers[3]  # we want the last one
print(last_report)
```

Python

```
# Your fix for Bug B:
# Hint: Python has a shortcut for accessing the last element of a list

# Error Type: IndexError
# Cause: report_numbers[3] is out of range
```

```
# Fix: use -1 to access the last element

report_numbers = ["01", "02", "03"]

last_report = report_numbers[-1]

print(last_report)
```

Python

```
# Bug C: run this, read the error, then fix it
filename = "report_01.docx"
clean_name = filename.Replace(".docx", "")
print(clean_name)
```

Python

```
# Your fix for Bug C:
# Error Type: AttributeError
# Cause: Replace() should be lowercase replace()
# Fix: use .replace()

filename = "report_01.docx"

clean_name = filename.replace(".docx", "")

print(clean_name)
```

Exercise 2: Find the silent bug

The program below is supposed to compute the average of three numbers entered by the user and tell them whether the average is above 50. It runs without errors but gives the wrong answer. Use **print(type(variable)** or **print(f"{variable = }")** to find and fix the bug.

Python

```
a = input("Enter a number: ")
```

```
b = input("Enter another number: ")
c = input("Enter one more number: ")

average = (a + b + c) / 3

if average > 50:
  print("The average is above 50")
else:
  print("The average is 50 or below")
```

Python

```
# Step 1: add type() print statements checks or self-documenting print statements
# to understand what a, b, c hold

# Then fix the program below

a = input("Enter a number: ")
b = input("Enter another number: ")
c = input("Enter one more number: ")

print(f"{a = }")
print(f"{b = }")
print(f"{c = }")

average = (int(a) + int(b) + int(c)) / 3

if average > 50:
  print("The average is above 50")
else:
  print("The average is 50 or below")
```

Exercise 3: Fix a broken file filter

The function below is supposed to return True if a filename ends with .csv and False otherwise. Several of the test cases return the wrong result. Use self-documenting expressions inside the function to understand what is going wrong, then fix it. **There are two bugs.**

Python

```python
def is_csv(filename: str) -> bool:
    """Returns True if the filename ends with .csv."""
    extension = filename[-3:]  # take the last 3 characters
    return extension == ".csv"

# Expected: True, True, True, False, False
print(is_csv("data.csv"))
print(is_csv("report.CSV"))       # uppercase extension
print(is_csv("Supplementary.csv"))
print(is_csv("notes.docx"))
print(is_csv("summary.pdf"))
```

Python

```python
# Step 1: add type() print statement or a self-documenting print inside the
# function to see what extension holds

# Try calling it with 'report.CSV' -- what does extension contain?
# Hint: count the characters in '.csv'

def is_csv_debug(filename: str) -> bool:
    extension = filename[-3:]
    # Add your self-documenting print here
    print(f"{extension = }")

    return extension == ".csv"
```

Python

```python
# Step 2: fix the function
# Hint: use .endswith() and .lower()

def is_csv_fixed(filename: str) -> bool:
    """Returns True if the filename ends with .csv (case-insensitive)."""
    return filename.lower().endswith(".csv")

# All five should now return the correct result
```

```
print(is_csv_fixed("data.csv"))        # True
print(is_csv_fixed("report.CSV"))      # True
print(is_csv_fixed("Supplementary.csv"))  # True
print(is_csv_fixed("notes.docx"))      # False
print(is_csv_fixed("summary.pdf"))     # False
```